

1M1B Green Internship

FINAL PROJECT SUBMISSION

(1M1B Green Internship, Batch 7)

- **Intern Name:** Asmita Pokharkar
- **College/University:** Lokmanya Tilak College of Engineering
- **Course/Year:** Btech - CSE AIML / 3 year
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I am a passionate AI & ML student with a strong interest in using data science for sustainability. My journey began with building small web applications and evolved into machine learning projects during internships at SmartIntern, Bharatcares and Personify. I have worked on water quality analysis using Python, developed an app that resells unsold food at a cheaper rate thus reducing food wastage, and have built multiple projects with sustainability and reducing waste in mind. I thrive on solving real-world problems with technology and believe that green skills are the future. Outside academics, I enjoy staying updated with tech trends, volunteering as a Campus Ambassador for IIT Bombay Techfest, and being part of the IIIC Cell to promote innovation. Currently working as a publicity head at my college ISTE Council.

2. Project Overview

- **Project Title:** AI-Based Water Demand Forecasting for Campus Operations
- **Project Theme (Waste, Water, Carbon Footprint, Energy):** water management
- **Group/Solo:** SOLO
- **Group name:** Not applicable
- **Group member's full official name:** Asmita Pokharkar
- **Problem Statement:** Campuses often struggle to manage daily water demand across hostels, canteens, academic blocks, and gardens. Sudden spikes cause shortages or over-pumping, leading to wasted water and electricity. A data-driven solution is needed to forecast demand and optimize pumping schedules.
- **Tools, Frameworks and Softwares used:** Python, TensorFlow (Keras), scikit-learn, pandas, NumPy, Matplotlib, Plotly, Google Colab, Tableau (for dashboard mockup), ipywidgets(colab)
- **Summary of Work Done:** I have developed an LSTM-based deep learning model to forecast next-day water demand for a campus zone (Hostel A) using 100 days of data (patterns). It was developed using colab for feasibility. The model uses features like temperature, rainfall, day of week, holiday status, and past demand. After training with a lookback window of 7 days, the model achieved a Mean Absolute Error of ~3,500 liters (~6% of average daily demand). The work demonstrates how Machine learning and AI can be applied to reduce water and energy waste in campus operations.

3. Links to Project Deliverables

SNO	Item	Google Drive Link
1	Master Google Drive Link	 1M1B Green Internship ...
2	Final PPT (PPT or PDF)	 AI-Based-Water-Deman...
3	Final Project Report / Concept Note(Word Document)	 1B1M project report
4	Tableau Dashboard (Direct link or link to the dashboard screenshot uploaded on google drive)	 1B1M
5	Project Photos (All the relevant photos of stakeholders interaction, implementation uploaded on a google drive folder)	 photos
6	Project Videos (Project Video compiled, uploaded on the google drive folder)	 project Video
7	Calculate the the final count of your LIP(Create a word document with a summary of your LIP engagements)	 PostAnalytics_AsmitaPo...
8	Any other supporting document (Raw data)	 campus_water_demand... for tableau Used synthetic data for predictive model

4. Impact Created

- **Quantitative Impact :**
- Energy Saved
- CO₂ Avoided - based on India grid factor 0.8 kg CO₂/kWh
- Water Saved - Reduced over-pumping
- Forecast Accuracy - leads to less wastage of resources
- **Qualitative Impact:**
- *Raised awareness*
- *Positive behaviour change*

5. Learnings from the Internship

I have understood how water and energy are interlinked; learned to quantify environmental impact (kWh, CO₂) and design solutions that align with sustainability goals and future needs . I have deepened expertise in time series analysis, feature engineering for LSTM models, and performance evaluation using accuracy factors. Also Gained hands-on experience with scaling, sequence creation, and hyperparameter tuning.

6. Challenges Faced

1. *Real campus data was not immediately accessible; I used synthetic data to simulate realistic patterns.*
2. *Working solo within 30 hours required careful prioritisation so I focused on one zone and used a streamlined pipeline.*

7. Acknowledgments

My mentors at SmartIntern and Edunet – for building my foundational datascience and AI-ML skills.

My college professors for giving much needed valuable insights .

Also my parents for their unconditional support and the 1B1M team for clear guidance.

8. Future Roadmap

Timeline	Planned Action
Next 1 month	Collect real historical water consumption data from my campus . Retrain the LSTM model with real data and validate its performance
Next 2 months	<i>Expand the model to all five campus zones.</i>
Next 6 months	<i>Make a working dashboard and deploying it on the internet .</i>
Next 12 months	<i>If possible converting it into a startup to create an impact.</i>

